Round-trip losses for varying gain medium profile widths

March 10, 2026

System

- Cavity length: $L = 150$ km
- Stability parameter: $g = 0$ (marginally stable)
- Wavelength: $\lambda = 1064$ nm
- Mirror diameters: $D = 1$ m
- Mirror radii of curvature: $R_c = 150$ km
- Fresnel number: $N_F = 1.566$
- Frame rotation speed: $\Omega = 0.001$ rad/s
- Frame acceleration: 1×10^{-5} m/s^2

Approach

For our ORACLE power beaming and photonic thrust applications, it is important to preserve as much laser power as possible.

Here we want to find the gain profile parameters such that the round-trip losses are minimized.

The round-trip loss fraction can be written as

$$loss_{RT} = 1 - T_{RT}, \tag{1}$$

where T_{RT} is the transmission over a round-trip.

In our cavity setup, we have a LHS mirror ("mirror 2") and a RHS mirror ("mirror 1"). A laser gain medium is also situated on the RHS.

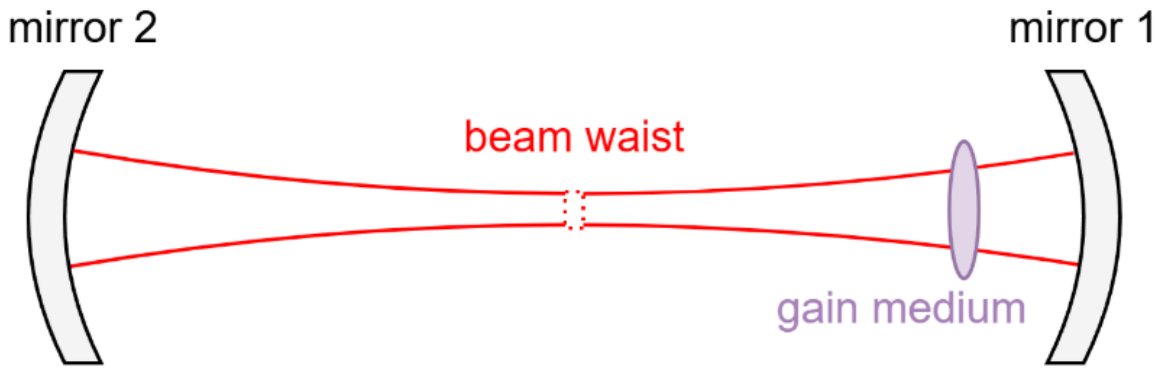


Figure 1: The laser cavity.

Each mirror has a transmission coefficient defined as the post-mirror (and therefore post-clipping) power divided by the pre-mirror power:

$$T = \frac{P_{after}}{P_{before}}. \tag{2}$$

And the corresponding mirror loss fraction is

$$loss = 1 - T. \tag{3}$$

Transmission and loss are computed before the gain element so that the mirror losses represent passive losses only.

Total round-trip loss is

$$loss_{RT} = 1 - T_1 T_2, \tag{4}$$

where T_1 is the transmission coefficient of mirror 1 and likewise, T_2 is the transmission coefficient of mirror 2.

This can be shown by considering initial laser power P_0. The power after mirror 1 is

$$P_1 = T_1 P_0. \tag{5}$$

The power after mirror 2 is

$$P_2 = T_2 P_1 = T_2 T_1 P_0. \tag{6}$$

Over one round-trip, the beam encounters each mirror once. So the round-trip transmission is

$$T_{RT} = T_1 T_2 \tag{7}$$

and the round-trip loss fraction is

$$loss_{RT} = 1 - T_1 T_2. \tag{8}$$

Results

Following the procedure outlined above, we find that the minimum round-trip loss occurs at $\sigma/R = 0.3$.

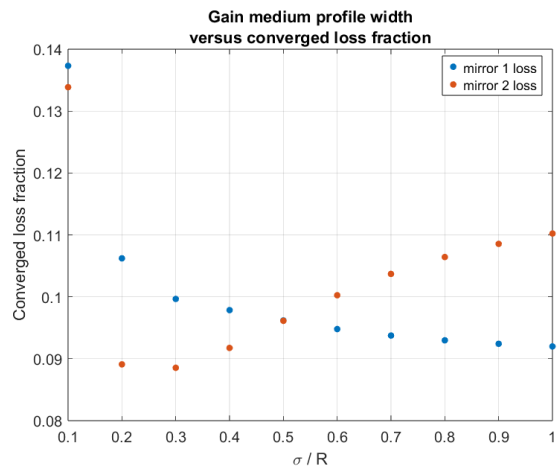


Figure 2: Loss fractions for each individual mirror. Loss at mirror 1 goes down as gain medium width increases, while loss at mirror 2 goes up with gain medium width.

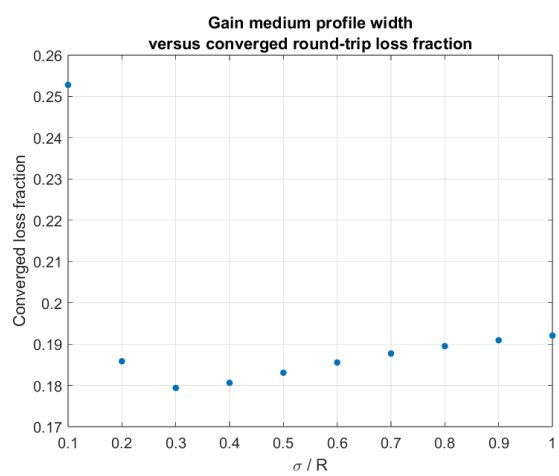


Figure 3: Combined round-trip loss fraction. The optimum occurs where the product of the individual mirror transmissions is largest, at $\sigma/R = 0.3$.

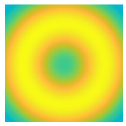
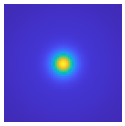
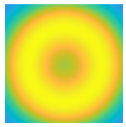
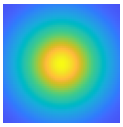
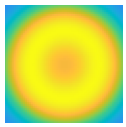
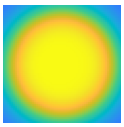
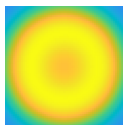
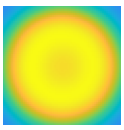
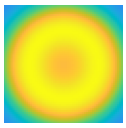
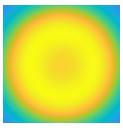
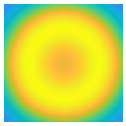
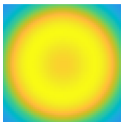
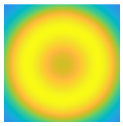
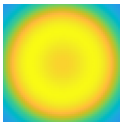
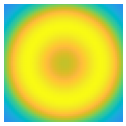
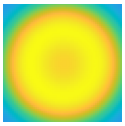
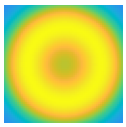
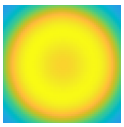
	LHS / Mirror 2	RHS / Mirror 1
$\sigma/R = 0.10$		
$\sigma/R = 0.20$		
$\sigma/R = 0.30$		
$\sigma/R = 0.40$		
$\sigma/R = 0.50$		
$\sigma/R = 0.60$		
$\sigma/R = 0.70$		
$\sigma/R = 0.80$		
$\sigma/R = 0.90$		
$\sigma/R = 1.00$		

Figure 4: Converged transverse laser modes at each mirror for varying gain profile widths σ/R.